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Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

Date / Revised: 15.08.2023 Version: 3.0
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Product: **Butadien**

(ID no. 30042222/SDS_GEN_DE/EN)

Date of print 12.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Butadien

Chemical name: 1,3-butadiene
INDEX-Number: 601-013-00-X
CAS Number: 106-99-0

REACH registration number: 01-2119471988-16, 01-2119471988-16-0029

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Chemical
Recommended use: Chemical

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

Flam. Gas 1	H220 Extremely flammable gas.
Press. Gas Liquefied gas	H280 Contains gas under pressure; may explode if heated.
Muta. 1B	H340 May cause genetic defects.
Carc. 1A	H350 May cause cancer.

According to BASF current knowledge and application of the criteria given in Annex I of Regulation (EC) No. 1272/2008, the following classification exceeding the classification given in Regulation (EC) No 1272/2008, Annex VI, Table 3.1 is required.

Flam. Gas 1A
Press. Gas Liquefied gas
Carc. 1A
Muta. 1B

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

H280	Contains gas under pressure; may explode if heated.
H220	Extremely flammable gas.
H350	May cause cancer.
H340	May cause genetic defects.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P201	Obtain special instructions before use.

Precautionary Statements (Response):

P308 + P313	IF exposed or concerned: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):

Restricted to professional users.

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
See section 12 - Results of PBT and vPvB assessment.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.
The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Chemical nature

1,3-butadiene

Content (W/W): >= 99,5 %	Flam. Gas 1
CAS Number: 106-99-0	Press. Gas Liquef. Gas
EC-Number: 203-450-8	Muta. 1B
INDEX-Number: 601-013-00-X	Carc. 1A
	H280, H220, H350, H340
Substance with EU occupational exposure limit	

stabilized with:
4-tert-Butylpyrocatechol

	Skin Corr./Irrit. 1B
CAS Number: 98-29-3	Eye Dam./Irrit. 1
EC-Number: 202-653-9	Skin Sens. 1
	Aquatic Chronic 2
	H317, H314, H411

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

3.2. Mixtures

Not applicable

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
carbon dioxide, dry powder

Unsuitable extinguishing media for safety reasons:
foam, water spray, water jet

Additional information:
Use extinguishing measures to suit surroundings.

5.2. Special hazards arising from the substance or mixture

Advice: Self-polymerization if overheated in a container.

Advice: Highly flammable. Vapours may form explosive mixture with air.

Advice: Shut off or stop released substance/product under safe conditions. Cool endangered containers with water-spray.

Advice: Burning produces harmful and toxic fumes.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Do not put fire out unless flow feeding it can be safely stopped. The substance/product forms flammable mixtures with air. Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings.

SECTION 6: Accidental Release Measures

Shut off or stop source of leak. Shut off or stop released substance/product under safe conditions.

6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing. Avoid all sources of ignition: heat, sparks, open flame. Wear respiratory protection if ventilation is inadequate.

Keep people away and stay on the upwind side.

Handle in accordance with good industrial hygiene and safety practice.

6.2. Environmental precautions

Contain contaminated water/firefighting water.

6.3. Methods and material for containment and cleaning up

Ensure adequate ventilation.

Suppress gases/vapours/mists with water spray jet.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Refill and handle product only in closed system. Handle in accordance with good industrial hygiene and safety practice. Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:
Avoid all sources of ignition: heat, sparks, open flame. Vapours may form explosive mixture with air.

Electrical devices must meet the specified temperature class.

Temperature class: T2 (Autoignition temperature >300 °C).

7.2. Conditions for safe storage, including any incompatibilities

No applicable information available.
Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.
Avoid all sources of ignition: heat, sparks, open flame.

Storage class according to TRGS 510 (originally VCI, Germany): (2A) Gases (except aerosol dispensers and lighters)

Storage stability:
The product is stabilized, the shelf life should be noted.

additives:
4-tert-Butylpyrocatechol (CAS Number: 98-29-3)

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

106-99-0: 1,3-butadiene

Tolerance Concentration (risk 4:1000): 5 mg/m3 ; 2 ppm (TRGS 910)
Acceptance concentration (risk 4:10000): 0,5 mg/m3 ; 0,2 ppm (TRGS 910)
Excursion factor (TRGS 910)
Ceiling limit value/factor: 8
Factor by which the average shift value (SMW) can be exceeded four times per shift during a maximum. period of 15 minutes each.
TWA value 2,2 mg/m3 ; 1 ppm (Directive 2004/37/EC)

PNEC

freshwater:
No PNEC derived, substance is a gas

marine water:
No PNEC derived, substance is a gas

intermittent release:
No PNEC derived, substance is a gas

sediment (freshwater):
No PNEC derived, substance is a gas

sediment (marine water):
No PNEC derived, substance is a gas

soil:
No PNEC derived, substance is a gas

STP:
No PNEC derived, substance is a gas

DNEL

worker:
Long-term exposure- systemic effects, Inhalation: 2,21 mg/m3, 1 ppm
The value corresponds to a DMEL.

consumer:
Long-term exposure- systemic effects, Inhalation: 0,0664 mg/m3
The value corresponds to a DMEL.

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if gases/vapours are formed. Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for gases/vapours of organic compounds (boiling point <65 °C, f.e. EN 14387 Type AX) Suitable respiratory protection for higher concentrations or long-term effect: Self-contained breathing apparatus.

Hand protection:
When there is a risk of frostbite from escaping gas, use thermally insulated gloves (EN 511).

Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
nitrile rubber (NBR) - 0.4 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness

Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
butyl rubber (butyl) - 0.7 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness

Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

Consider the risk management measures as outlined in the exposure scenario.

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Avoid contact with the skin, eyes and clothing. Ensure adequate ventilation. Avoid inhalation of vapour. Remove

contaminated clothing immediately and dispose of safely. At the end of the shift the skin should be cleaned and skin-care agents applied.

Environmental exposure controls

Suitable risk management measures should be in place.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	gaseous	
Form:	pressurised liquified gas	
Colour:	colourless	
Odour:	aromatic	
Odour threshold:	not determined	
Melting point:	-108,9 °C (1.013 hPa) Literature data.	
Boiling point:	-4,41 °C (1.013 hPa) Literature data.	
Flammability:	Extremely flammable.	(derived from explosion limits)
Lower explosion limit:	2,0 %(V) Literature data.	
Upper explosion limit:	12 %(V) Literature data.	
Flash point:	-76,1 °C Literature data.	(other, other)
Auto-ignition temperature:	420 °C Literature data.	
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.	
pH value:	The substance does not dissociate.	
Viscosity, kinematic:	not applicable, the product is a gas	
Viscosity, dynamic:	0,149 mPa.s (20 °C)	
Thixotropy:	not thixotropic	
Solubility in water:	Literature data. 0,735 g/l (20 °C)	
Solubility (qualitative) solvent(s):	organic solvents soluble	
Partitioning coefficient n-octanol/water (log Kow):	1,99 (25 °C; pH value: 7)	

Vapour pressure:	2.170 hPa (16,85 °C) Literature data. 2.550 hPa (21,85 °C) Literature data.	
Relative density:	0,0023 (15 °C)	(calculated)
Density:	0,0023 g/cm3 (15 °C, 1.013 hPa) Study scientifically not justified.	(calculated)
Relative vapour density (air):	1,9322 Literature data., Heavier than air.	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Oxidizing properties

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Gases under pressure

Critical temperature/pressure: 152,0 °C
Literature data.

Pyrophoric properties

Self-ignition temperature: Test type: Spontaneous self-ignition at room-temperature.

Based on its structural properties the product is not classified as self-igniting.

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating.

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases:
Forms no flammable gases in the presence of water.

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

Radioactivity:		not radioactive for transport purposes
pKA:	The substance does not dissociate.	
Adsorption/water - soil:	KOC: 39,6; log KOC: 1,6	(calculated)
Surface tension:	Based on chemical structure, surface activity is not to be expected.	
Molar mass:	54,09 g/mol	
SAPT-Temperature:	50 °C insulated tank, 20 m3	
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:	No corrosive effect on metal.	
Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated. The product is stabilized against spontaneous polymerization prior to despatch.

Peroxides:	The product/the substance has a tendency towards the formation of peroxide.

10.3. Possibility of hazardous reactions

Violent reaction under influence of oxidizing agents.
The product is stabilized against spontaneous polymerization prior to despatch.

10.4. Conditions to avoid

Avoid all sources of ignition: heat, sparks, open flame. Avoid direct sunlight. Heating may cause pressure build-up and possible rupture of the container.

10.5. Incompatible materials

Substances to avoid:
strong oxidizing agents

10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Virtually nontoxic by inhalation.

Experimental/calculated data:

(oral):Study technically not feasible.
LC50 rat (by inhalation): 285 mg/l 4 h
A gas was tested.
(dermal):Study technically not feasible.

Irritation

Assessment of irritating effects:

Contact with liquid may cause frostbite. The substance is explosive and gaseous at room temperature and pressure. Testing for this particular endpoint is technically not feasible and/or this endpoint does not represent a relevant exposure scenario.

Experimental/calculated data:

Skin corrosion/irritation

: Study technically not feasible.

Serious eye damage/irritation

: Study technically not feasible.

Respiratory/Skin sensitization

Assessment of sensitization:

There is no evidence of a skin-sensitizing potential. The substance is explosive and gaseous at room temperature and pressure. Testing for this particular endpoint is technically not feasible and/or this endpoint does not represent a relevant exposure scenario.

Experimental/calculated data:

Study technically not feasible.

Germ cell mutagenicity

Assessment of mutagenicity:

Capable of causing genetic defects.

Carcinogenicity

Assessment of carcinogenicity:
The substance caused cancer in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:
Animal studies gave no indication of a fertility impairing effect at doses which were not toxic to the parental animals.

Developmental toxicity

Assessment of teratogenicity:

Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated inhalation exposure to large quantities may affect certain organs.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms.
The product has not been tested. The statement has been derived from the structure of the product.

Toxicity to fish:
LC50 (96 h) 45 mg/l, Pimephales promelas (calculated, other)

Aquatic invertebrates:
EC50 (48 h) 33 mg/l, Daphnia magna (calculated, other)

Aquatic plants:
EC50 (72 h) 33 mg/l (growth rate), algae (calculated, other)

Microorganisms/Effect on activated sludge:
No data available.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:
No data available.

Assessment of terrestrial toxicity:
No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Elimination information:
No data available.

Assessment of stability in water:
No data available.

Information on Stability in Water (Hydrolysis):
No data available.

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Bioaccumulation potential:
No data available.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will rapidly evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is possible.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Dispose of in accordance with national, state and local regulations.

Contaminated packaging:
Disposal must be made according to official regulations.

SECTION 14: Transport Information

Land transport

ADR

UN number or ID number:	UN1010
UN proper shipping name:	BUTADIENES, STABILIZED
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	no
Special precautions for user:	Tunnel code: B/D

RID

UN number or ID number:	UN1010
UN proper shipping name:	BUTADIENES, STABILIZED
Transport hazard class(es):	2.1, 13
Packing group:	Not applicable
Environmental hazards:	no
Special precautions for user:	Shunting label: 13

Inland waterway transport

ADN

UN number or ID number:	UN1010
UN proper shipping name:	BUTADIENES, STABILIZED
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	no
Special precautions for user:	None known

Transport in inland waterway vessel

UN number or ID number:	UN1010
UN proper shipping name:	BUTADIENES (1,3-BUTADIENE), STABILIZED
Transport hazard class(es):	2.1, INST, CMR
Packing group:	Not applicable
Environmental hazards:	no
Type of inland waterway vessel:	G

Cargo tank design:	1
Cargo tank type:	1

Sea transport

IMDG

UN number or ID number:	UN 1010
UN proper shipping name:	BUTADIENES, STABILIZED
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	EmS: F-D; S-U

Air transport

IATA/ICAO

UN number or ID number:	UN 1010
UN proper shipping name:	BUTADIENES, STABILIZED
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards

See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Regulation:	IGC-Code
Product name:	butadiene
Ship Type:	2G/2GP

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Chemical Prohibition Ordinance (DE): Annex 2
Restriction Type: Restricted substance

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 40, 28, 29, 75, 40, 28, 29, 75

Hazardous Incident Ordinance (Germany):
List entry in regulation: 1.2.2
Listed in above regulation: Liquefied flammable gases, Category 1 or 2 (including LPG) and natural gas

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
List entry in regulation: P2
Listed in above regulation: Liquefied flammable gases, Category 1 or 2 (including LPG) and natural gas

Classification according to 'TA-Luft' (Germany):

5.2.7.1.1 class III: Carcinogenic substances class III

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (3) Strongly water polluting. ID-No.: 218

Regulation on prohibitions and restrictions on the marketing of dangerous substances, preparations and goods in accordance with the chemical law (Germany)
Observe TRGS 910 on cmr substances (German Technical Rule for Hazardous Substances)
List of carcinogenic, mutagenic or reprotoxic substances (TRGS 905)
German Regulation TA Luft (Technical Instruction on Air Quality Control, i.e. first Directive to the Federal Immission Control Ordinance)
Law on the Protection of Working Youth
The Maternity Protection Act needs to be considered.

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Flam. Gas 1A
Carc. 1A
Muta. 1B
Press. Gas Liquefied gas

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Flam. Gas	Flammable gases
Press. Gas	Gases under pressure
Muta.	Germ cell mutagenicity
Carc.	Carcinogenicity
H280	Contains gas under pressure; may explode if heated.
H220	Extremely flammable gas.
H350	May cause cancer.
H340	May cause genetic defects.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent,

Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.

Annex: Exposure Scenarios

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1. Manufacture of substance
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2. Formulation
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3. Use as laboratory reagent/agent
IS; ERC4; PROC15

4. Polymer production
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5. Use as a Fuel
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6. Polymer processing
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7. Use of a substance as an intermediate
IS; ERC6a; PROC1, PROC2, PROC3, PROC8a, PROC8b, PROC9, PROC15

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IS; ERC6d; PROC1, PROC2, PROC3, PROC6, PROC8a, PROC8b, PROC9, PROC14, PROC15

9. Polymer processing, (use in professional settings)
PW; ERC8a; PROC1, PROC2, PROC6, PROC8a, PROC8b

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1. Short title of exposure scenario

Manufacture of substance
M; ERC1; PROC1, PROC2, PROC3, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC1: Manufacture of the substance As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. General exposures (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN	

ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial	

enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills	

immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general	Effectiveness: 30 %

ventilation (not less than 3 - 5 air changes per hour)	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained

	batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	

Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial	

enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	

For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin	

contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use dry break couplings for material transfer	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular	

inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa

Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	

Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use dry break couplings for material transfer	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid

Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills	

immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of	

equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1269 mg/m³
Risk Characterization Ratio (RCR)	0,509902
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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2. Short title of exposure scenario

Formulation
F; ERC2; PROC1, PROC2, PROC3, PROC5, PROC8a, PROC8b, PROC9, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture As no environmental hazard was identified no environmental-related exposure assessment and risk

	characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. General exposures (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin	

contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) with sample collection With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) with sample collection With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³

Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) with sample collection With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or	

equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	

For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with	

substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Batch processes at elevated temperatures Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	2,0284 mg/m³
Risk Characterization Ratio (RCR)	0,917824
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Batch processes at elevated temperatures Use domain: industrial
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	2,0284 mg/m³
Risk Characterization Ratio (RCR)	0,917824
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or	

equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment

	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills	

immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 95 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1832 mg/m³
Risk Characterization Ratio (RCR)	0,535397
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance.	
Provide extract ventilation to points where emissions occur (LEV).	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1832 mg/m³
Risk Characterization Ratio (RCR)	0,535397
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities (manual) Transfer from/pouring from containers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract	

ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities (manual) Transfer from/pouring from containers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	

Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with	

substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid

Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment

	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Drum and small package filling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air	Effectiveness: 70 %

changes per hour)	
Handle substance within closed system.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Drum and small package filling Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Handle substance within closed system.	Effectiveness: 90 %
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	

In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	

Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing).

	Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	

Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Tabletting, compression, extrusion, pelletisation, granulation Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract	

ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated

	temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario

Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	1,1269 mg/m³
Risk Characterization Ratio (RCR)	0,509902
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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3. Short title of exposure scenario

Use as laboratory reagent/agent
IS; ERC4; PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities small scale Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1269 mg/m³
Risk Characterization Ratio (RCR)	0,509902
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities small scale Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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4. Short title of exposure scenario

Polymer production
IS; ERC6c; PROC1, PROC2, PROC3, PROC6, PROC8a, PROC8b, PROC14

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6c: Use of monomer in polymerisation processes at industrial site (inclusion or not into/onto article) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. General exposures (closed systems) Continuous process no sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Polymerisation (bulk and batch) Continuous process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene

	Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³

Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Polymerisation (bulk and batch) Continuous process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must	

be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor

	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additivation and stabilisation Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin	

contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additivation and stabilisation Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Bulk transfer Continuous process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Bulk transfer Continuous process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must	

be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Finishing operations Batch process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Finishing operations Batch process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with	

substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,9466 mg/m³
Risk Characterization Ratio (RCR)	0,428318
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

	Worker - inhalation, long-term - systemic
Exposure estimate	0,9466 mg/m³
Risk Characterization Ratio (RCR)	0,428318
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract	

ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Polymerisation (bulk and batch) Batch process with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Pelletizing Extrusion and masterbatching Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as	

soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Pelletizing Extrusion and masterbatching Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance.	

Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer transport with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or	Effectiveness: 70 %

controlled ventilation (5 to 10 air changes per hour)	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer transport with sample collection Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer transport with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 95 %
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must	

be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer transport with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated

	temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities

	Pelletisation and pellet screening (open systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Pelletisation and pellet screening (open systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Ensure material transfers are under containment or extract ventilation	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use	

suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities transport with sample collection Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	

Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities transport with sample collection Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Pelletizing Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must	

be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Pelletizing Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures

Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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5. Short title of exposure scenario

Use as a Fuel
IS; ERC7; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC16

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC7: Use of functional fluid at industrial site As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. General exposures (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or	

equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Storage Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed

	continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to	

prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Handle substance within closed system.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular	

inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid

Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment

	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves.,	

Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Batch process Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures

Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical

	industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Batch process Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Handle substance within closed system.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin	

contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (open systems) (closed systems) Batch process Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,4057 mg/m³
Risk Characterization Ratio (RCR)	0,183565
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (open systems) (closed systems) Batch process Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	0,4057 mg/m³
Risk Characterization Ratio (RCR)	0,183565
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC4: Chemical production where opportunity for exposure arises Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use	

suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,8114 mg/m³
Risk Characterization Ratio (RCR)	0,367129
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC4: Chemical production where opportunity for exposure arises Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	

Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,8114 mg/m³
Risk Characterization Ratio (RCR)	0,367129
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Disposal of wastes (Disposal/transfer) Use domain: industrial
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	2,0284 mg/m³
Risk Characterization Ratio (RCR)	0,917824
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Disposal of wastes (Disposal/transfer) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Wear suitable respiratory protection.	Effectiveness: 90 %
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves.,	

Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	2,0284 mg/m³
Risk Characterization Ratio (RCR)	0,917824
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8256 mg/m³
Risk Characterization Ratio (RCR)	0,826041
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular	

inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8256 mg/m³
Risk Characterization Ratio (RCR)	0,826041
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa

Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8256 mg/m³
Risk Characterization Ratio (RCR)	0,826041
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Transfer via enclosed lines	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8256 mg/m³
Risk Characterization Ratio (RCR)	0,826041
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC16: Use of fuels General exposures (open systems) (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 25 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air	Effectiveness: 30 %

changes per hour)	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,4733 mg/m³
Risk Characterization Ratio (RCR)	0,214159
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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6. Short title of exposure scenario

Polymer processing
IS; ERC6d; PROC1, PROC2, PROC3, PROC6, PROC8a, PROC8b, PROC9, PROC13, PROC14

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6d: Use of reactive process regulators in polymerisation processes at industrial site (inclusion or not into/onto article) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Bulk transfer (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Bulk weighing (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance.	

Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk transfer (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,5634 mg/m³
Risk Characterization Ratio (RCR)	0,254951
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk transfer (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Handle substance within closed system.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,5634 mg/m³
Risk Characterization Ratio (RCR)	0,254951
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk weighing With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,3944 mg/m³
Risk Characterization Ratio (RCR)	0,178466
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk weighing With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene

	Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,3944 mg/m³
Risk Characterization Ratio (RCR)	0,178466
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal -	

This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additive premixing (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week

Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additive premixing (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as	

soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Calendering (including Banburys) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of	

equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Calendering (including Banburys) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to	

prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance	268333,1875 Pa

during use	
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Ensure material transfers are under containment or extract ventilation	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide	

specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6903 mg/m³
Risk Characterization Ratio (RCR)	0,764853
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Drum/Batch transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	

Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1832 mg/m³
Risk Characterization Ratio (RCR)	0,535397
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Bulk transfer Small package filling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8932 mg/m³
Risk Characterization Ratio (RCR)	0,856635
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	

For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Bulk transfer Small package filling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as	

soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,8932 mg/m³
Risk Characterization Ratio (RCR)	0,856635
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Small scale weighing Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular	

inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Small scale weighing Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa

Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC13: Treatment of articles by dipping and pouring. Production of articles by dipping and pouring Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC13: Treatment of articles by dipping and pouring. Production of articles by dipping and pouring Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Extrusion and masterbatching Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Extrusion and masterbatching Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	

In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Injection moulding (of articles) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	

Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Injection moulding (of articles) Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	

Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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7. Short title of exposure scenario

Use of a substance as an intermediate
IS; ERC6a; PROC1, PROC2, PROC3, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6a: Use of intermediate As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. General exposures (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario

Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide	

specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions General exposures (closed systems) With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons.	

Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use	

suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition General exposures (closed systems) Use in contained batch processes Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated

	temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide	

specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of	

equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Cleaning Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as	

soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular	

inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa

Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use dry break couplings for material transfer	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Bulk transfer (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Use dry break couplings for material transfer	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance.	

Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Process sampling Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	15 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Sample via a closed loop or other system to avoid exposure.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,3523 mg/m³
Risk Characterization Ratio (RCR)	0,611882
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1269 mg/m³
Risk Characterization Ratio (RCR)	0,509902
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

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8. Short title of exposure scenario

Use in rubber production and processing
IS; ERC6d; PROC1, PROC2, PROC3, PROC6, PROC8a, PROC8b, PROC9, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC6d: Use of reactive process regulators in polymerisation processes at industrial site (inclusion or not into/onto article) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Bulk weighing (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance.	

Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0158 mg/m³
Risk Characterization Ratio (RCR)	0,007139
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Material transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,0225 mg/m³
Risk Characterization Ratio (RCR)	0,010198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk weighing With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk weighing With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Material transfers With occasional controlled exposure. Use domain: industrial
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Material transfers With occasional controlled exposure. Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use	

suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additive premixing Batch process (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated

	temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario

Use descriptors covered	PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition Additive premixing Batch process (closed systems) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as	

soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,6761 mg/m³
Risk Characterization Ratio (RCR)	0,305941
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Calendering (including Banburys) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of	

equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Calendering (including Banburys) Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C

Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Cooling cured articles Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	

Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Cooling cured articles Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to	

equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Vulcanisation Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %

Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Vulcanisation Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves.,	

Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %

Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker

	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	

In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	

Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Dedicated facility Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent	

exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Material transfers Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air	Effectiveness: 70 %

changes per hour)	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,8114 mg/m³
Risk Characterization Ratio (RCR)	0,367129
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Material transfers Use domain: industrial
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Transfer via enclosed lines	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,8114 mg/m³
Risk Characterization Ratio (RCR)	0,367129

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Small scale weighing Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves.,	

Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6227 mg/m³
Risk Characterization Ratio (RCR)	0,734259
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Small scale weighing Dedicated facility Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 5 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or	Effectiveness: 70 %

controlled ventilation (5 to 10 air changes per hour)	
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,6227 mg/m³
Risk Characterization Ratio (RCR)	0,734259
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Pressing uncured rubber blanks Use domain: industrial

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC14: Tabletting, compression, extrusion, pelletisation, granulation Pressing uncured rubber blanks Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	240 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general or controlled ventilation (5 to 10 air changes per hour)	Effectiveness: 70 %
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal -	

This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,0142 mg/m³
Risk Characterization Ratio (RCR)	0,458912
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent. Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures

Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC15: Use a laboratory reagent.

	Laboratory activities Use domain: industrial
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,1269 mg/m³

Risk Characterization Ratio (RCR)	0,509902
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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9. Short title of exposure scenario

Polymer processing, (use in professional settings)
PW; ERC8a; PROC1, PROC2, PROC6, PROC8a, PROC8b

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC8a: Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor) As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.
Operational conditions	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Bulk transfer (closed systems) Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	

Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,2254 mg/m³
Risk Characterization Ratio (RCR)	0,10198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC1: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. Storage Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene

	Content: >= 0 % - <= 100 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,2254 mg/m³
Risk Characterization Ratio (RCR)	0,10198
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal

Additional good practice advice
Where appropriate, replacement of task by automated and/or closed processes.
Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk transfer (closed systems) With occasional controlled exposure. Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills	

immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Bulk transfer (closed systems) With occasional controlled exposure. Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %

Ensure material transfers are under containment or extract ventilation	Effectiveness: 80 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: professional

Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	

Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC2: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions Storage With occasional controlled exposure. Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Handle substance within closed system.	Effectiveness: 80 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or	

equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Injection moulding (of articles) Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor

	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 95 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	

Guidance to Downstream Users
For scaling see: http://www.ecetoc.org/tra

Contributing exposure scenario	
Use descriptors covered	PROC6: Calendering operations Injection moulding (of articles) Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings.	Effectiveness: 95 %
Handle substance within a predominantly closed system provided with extract ventilation.	Effectiveness: 80 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable	

respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Wear suitable respiratory protection.	Effectiveness: 90 %
Provide a good standard of general ventilation (not less than 3 - 5 air	Effectiveness: 30 %

changes per hour)	
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Equipment maintenance Use domain: professional
Operational conditions	

Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	480 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Wear suitable respiratory protection.	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic

Exposure estimate	1,5776 mg/m³
Risk Characterization Ratio (RCR)	0,713863
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Local exhaust ventilation	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour)	Effectiveness: 30 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use	

suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities Material transfers Use domain: professional
Operational conditions	
Concentration of the substance	1,3-butadiene Content: >= 0 % - <= 1 %
Physical state	liquid
Vapour pressure of the substance during use	268333,1875 Pa
Process temperature	20 °C
Duration and Frequency of activity	60 min 5 days per week
Indoor/Outdoor	Indoor
	Operation is carried out at ambient or elevated temperatures
Risk Management Measures	
Provide a good standard of general	Effectiveness: 30 %

ventilation (not less than 3 - 5 air changes per hour)	
Ensure material transfers are under containment or extract ventilation	Effectiveness: 90 %
Drain down and flush system prior to equipment break-in or maintenance. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Regular inspection and maintenance of equipment and machines.	
Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Disposal - This material and its container must be disposed of in a safe manner.	
In case of potential exposure:, Use suitable chemically resistant gloves., Wear suitable coveralls to prevent exposure to the skin., Wear suitable respiratory protection., Clear spills immediately, Consider the need for risk based health surveillance.	
Avoid frequent and direct contact with substance. Clean up contamination as soon as they occur. Wash off any skin contamination immediately. Provide specific employee training to prevent/minimize exposures.	
Wear suitable gloves tested to EN ISO 374-1.	
Exposure estimate and reference to its source	
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0,7888 mg/m³
Risk Characterization Ratio (RCR)	0,356931
Assessment method	EASY TRA v6.0, ECETOC TRA v3.0, Worker
Assessment method	Qualitative assessment
	Worker - dermal
Additional good practice advice	
Where appropriate, replacement of task by automated and/or closed processes.	
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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